

Expanding School Enrollment by Subsidizing Private Schools: Lessons from Bogotá

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I. Introduction

In many countries government policies that use tax revenues to pay all or part of the cost of educating children in private schools have blurred the distinction between public and private schools. For example, Chile and New Zealand have implemented educational voucher programs to stimulate competition among schools.¹ The government of the Netherlands pays the costs of educating children at schools run by religious organizations.² Recently, some countries have introduced policies of subsidizing private schools as a way of increasing supply.³ For example, part of the strategy the city of Bogotá, Colombia, is using to meet its commitment to universal access to primary schooling is to subsidize private schools that enroll low-income students.

The success of these private sector strategies assumes that participating private schools will provide a high-quality education to their students and will do so without reducing the quality of public schools.⁴ Understanding whether this assumption is valid requires investigation of the manner in which public schools and private schools “produce” student achievement, especially because, in many settings, the student body compositions, teacher characteristics, and class sizes of public schools and private schools differ markedly.⁵ If these factors play a role in determining children’s skills, then these dif-

¹ Edward B. Fiske and Helen F. Ladd, *When Schools Compete: A Cautionary Tale* (Washington, DC: Brookings Institution, 2000); Chang-Tai Hsieh and Miguel Urquiola, “When Schools Compete, How Do They Compete? An Assessment of Chile’s Nationwide School Voucher Program,” NBER Working Paper no. 10008 (National Bureau of Economic Research, Cambridge, MA, 2003).

² Estelle James, “Public Subsidies for Private and Public Education: The Dutch Case.” In *Private Education: Studies in Choice and Public Policy*, ed. D. C. Levy, 113–37 (Oxford: Oxford University Press, 1986).

³ Indeed, Estelle James (“Why Do Different Countries Choose a Different Public-Private Mix of Educational Services?” *Journal of Human Resources* 28 [Summer 1993]: 571–92) finds that countries with the largest percentage of private school enrollments are those with excess demand for schooling.

⁴ There is some evidence that private schools are more cost-effective than private schools. For example, using data from Indonesia, Estelle James, Elizabeth M. King, and Ace Suryadi (“Finance, Management, and Costs of Public and Private Schools in Indonesia,” *Economics of Education Review* 15 [October 1996]: 387–98) find that the cost per student of achieving a given level of academic performance is lower in private schools than in public schools.

⁵ Patrick McEwan and Martin Carnoy, “The Effectiveness and Efficiency of Private Schools in Chile’s Voucher System,” *Educational Evaluation and Policy Analysis* 22 (Autumn 2000): 213–39.

ferences may compromise the success of private sector strategies in providing a high-quality education to the children they are meant to help.

Understanding the effects of teacher quality, class size, and peer group composition on school success is relevant to the design of public policies regarding the rules under which private schools operate. For example, the larger the role of peer group composition in determining students' achievement, the more important it is to design public policies that prevent the concentration of low-income children in particular schools.

Disentangling the relative contributions of class size, peer group composition, and teacher quality is not possible in typical data sets in which each teacher works with only one group of children. In this study, however, we make use of an unusual property of a data set from Bogotá—whereby some teachers teach math to more than one group of students—to determine the roles of teacher quality, peer group composition, and class size. We show that all three have effects on student achievement. We also show that the average attributes of public schools and private schools differ markedly in dimensions that affect student achievement. Finally, we show that differences in student achievement among schools in the public sector, and especially among schools in the private sector, are much larger than the average difference between the achievement of children attending public schools and those attending private schools, net of family background influences. The implication for parents is that the decision to use public or private schools is much less important than the choice of school within either sector. The implication for governments is that the details of subsidy programs are extremely important.

Section II provides a brief summary of Bogotá's recent strategy of subsidizing private schools. Section III discusses previous research on the effects of class size, peer group composition, and teacher characteristics, particularly in Latin America, with a special focus on the studies that have used the most plausible research designs. Section IV presents the data, sample, measures, and analytical strategies that we use to answer our research questions. Section V presents our findings, and Section VI concludes with a discussion of the implications of these findings for school policy in Bogotá and in other settings in which public funds are provided to private schools.

II. The Bogotá Setting

Since the 1950s, Bogotá has experienced a sixfold growth in population as migrants have streamed into the city in search of better economic opportunities or a safe refuge from the armed conflict that has ravaged large parts of the country. Providing access to schooling for the growing population has been a major challenge for the Bogotá government. In 1997 when a new mayoral administration came into office, it found that more than 140,000 children in the city from the lowest income level were not attending school.

Of those students from low-income families who were in school, close to one-third were enrolled in private unsubsidized schools of dubious quality.⁶

Committed to providing access to publicly funded schooling for all school-age children, the Department of Education (*Secretaría de Educación del Distrito*—SED) has used a two-pronged strategy to meet the demand for primary schooling. One prong involves expansion of the public sector, by reassigning teachers from administrative posts to classrooms, constructing or rehabilitating classrooms in run-down schools, and increasing class sizes.⁷ A related policy is to give students from low-income families preferential access to the limited number of places available in public schools.

The second prong involves subsidizing private schools that enroll low-income students. As a result, in the year 2000, 53 percent of students in the city were enrolled in the city's 2,900 private schools.⁸ For-profit schools, typically started by individuals, constitute 90 percent of all private schools in the city. The second most common type are religious schools, which make up 8 percent of the total and are usually either subsidized or “contracted” by the government to provide educational services in low-income areas. Finally, cooperative schools are established by neighborhood organizations and derive their funding from both tuition and public subsidies.

Public and private school sectors in Bogotá differ in a number of ways, particularly with respect to student composition, teacher qualifications, and class size. Consequently, even though SED's strategy of subsidizing private schools may have increased access to primary schooling in the city, the schooling experiences of low-income students attending schools in the two sectors may be quite different.

Private schools in Bogotá may set their own student admission policies, which are based mostly on academic criteria and/or on families' ability to pay.⁹ Although private schools in Bogotá enroll students from the whole range of the socioeconomic spectrum, on average they serve a more affluent segment of the population than do public schools. For example, 43 percent of the private schools in our sample enroll primarily students from the lowest two (of six) socioeconomic strata, while 10 percent enroll primarily students

⁶ Secretaría de Educación del Distrito, *Informe de Gestión: 1998–2000* (Bogotá: Secretaría de Educación del Distrito, 2000).

⁷ The SED established a minimum of 35 students per class in grades 1–5 and of 40 in grades 6–11.

⁸ In comparison, only 21 percent of all pre-university-level students in the country were enrolled in private schools in the same year; from Secretaría de Educación del Distrito, *Boletín Estadístico: Versión Borrador* (Bogotá: Secretaría de Educación del Distrito, 2001).

⁹ The yearly tuition fee for private schools in Bogotá in the 2002 school year ranged from \$30 to \$2,400, with an average of \$285; Secretaría de Educación del Distrito, *Estadísticas Educativas, 1998–2003* (Bogotá: Secretaría de Educación Distrital, 2003). This is considerably less than the \$439 average expenditure per student in public schools. However, these figures are not completely comparable, because the public-sector expenditure figure includes the costs of school supplies, transportation, and nutritional supplement, while the private school tuitions do not. Also, many private schools impose additional fees for school activities and expect parents to make “voluntary” donations (*bonos*).

from the highest three strata.¹⁰ In comparison, 75 percent of public schools in our sample enroll students from mainly the lowest two strata, and none enrolls students from the upper income strata.

Public and private schools also differ in the characteristics of their teachers. Most public primary school teachers in Bogotá have 4-year degrees in education and postgraduate training.¹¹ As a result of relatively low turnover (3 percent annually) among Bogotá public school teachers and restrictions on the opening of new teaching posts in the public sector, the public school teaching force is mostly middle-aged and experienced.¹² In contrast, private school teachers in Bogotá have much less teaching experience and fewer formal educational credentials than public school teachers.¹³

There are a number of concerns about the impact that the recent reforms aimed at increasing access have had on the quality of education provided to students. One stems from the increase in class sizes in public sector schools. A second is that schools may be becoming more segregated by socioeconomic status (SES) as students from the lowest income groups are given preferential assignment to public and private subsidized schools, leaving middle-class children to attend unsubsidized private schools. A third is that, as the city increases its subsidies program, some private schools accustomed to serving a relatively affluent population are increasingly serving children from low-income families. Serving these children poses new challenges that are difficult to meet, especially by schools that contain costs by hiring teachers with little pedagogical training or teaching experience.

If class size, peer group composition, and teacher qualifications affect student achievement, the reforms aimed at increasing access may jeopardize quality in both sectors. This is why we explore the effects these factors have on student achievement and why we examine the roles they play in explaining the relative effectiveness of public and private schools.

Before describing our approach to answering these questions, we review

¹⁰ Bogotá has a six-level system of socioeconomic stratification, with six as the highest and one as the lowest. Roughly, *estratos* 1 and 2 correspond to families earning up to two times the minimum wage (the 2003 minimum wage is approximately US\$100/month). *Estratos* 4 and 5 families earn approximately five to ten times the minimum wage.

¹¹ Based on data from A. Sarmiento, B. L. Caro, J. I. González, E. Castaño, and J. Espinosa, *Evaluación de la calidad de la educación primaria en Santa Fe de Bogotá 1998: Factores asociados al logro; Informe final versión definitiva* (Bogotá: Departamento Nacional de Planeación, Secretaría de Educación de Bogotá, Corporación Mixta para el Desarrollo de la Educación Básica, 1999).

¹² New regulations that came into effect in 2002 introduce performance accountability mechanisms for promotions and reduce the weight of academic credentials and experience in determining promotions. These changes are likely to bring about changes in the city's teaching force over the coming years.

¹³ Sarmiento et al., *Evaluación de la calidad*; Alejandro Gaviria and Jorge Hugo Barrientos, "Determinantes de la calidad de la educación en Colombia," *Planeación y Desarrollo* 32 (Autumn 2001): 339–86. A 1995 constitutional court ruling (related to article 198 of the 1991 constitution) required private schools to pay teachers according to the public teacher salary scale, though private schools still have greater flexibility than public schools in hiring teachers with modest credentials and in dismissing teachers at will.

research from other countries. We limit our review to studies that have employed compelling strategies for estimating the unbiased causal impacts of class size, peer group composition, and teacher quality on student achievement and the relative effectiveness of public and private schools.

III. Prior Research

A substantial literature examines the effects of class size, peer group composition, and teacher quality on student achievement, both in the United States and in developing countries.¹⁴ Most studies are problematic, however, because they lack compelling strategies to account for the nonrandom assignment of teachers and students to classrooms. For instance, most studies of the effects of class size on student achievement do not adequately account for the assignment of weaker students (or stronger students) to smaller classes. As a result, the estimates of class size effects are likely to be biased. Similarly, it is likely that highly motivated parents choose their child's school based on the characteristics of its students. Consequently, in many studies, estimates of the effect of peer group composition may be biased because researchers do not adequately control for the unobserved influences of parental motivation on children's achievement. Following this same logic, if better teachers are assigned to more able students (or vice versa), then the estimated effects of teacher characteristics on student achievement will also be biased by the omission of these unobserved effects from the statistical models.

It follows that a powerful strategy for obtaining unbiased estimates of the causal effects of class size, peer group composition, and teacher characteristics on achievement is to randomly assign teachers and students to classrooms. Randomized controlled trials are expensive, however, and remain few in number.¹⁵ The next best alternative is to learn from "natural experiments" in which a policy quirk creates exogenous variation in the assignment of students to classrooms.

Miguel Urquiola, for example, estimates the effects of class size by exploiting the results of a policy rule in Bolivia: when class enrollment size

¹⁴ For an older review of research in developing countries, see Bruce Fuller, "What School Factors Raise Achievement in the Third World?" *Review of Educational Research* 57 (Autumn 1987): 255–92.

¹⁵ The only class size experiment is Project STAR in Tennessee; see A. B. Krueger, "Experimental Estimates of Education Production Functions," *Quarterly Journal of Economics* 114 (May 1999): 497–532. In terms of peer effects, the Moving to Opportunity project randomly awards housing vouchers to residents of low-income neighborhoods in various cities in the United States. For results from Boston, see Lawrence F. Katz, Jeffrey R. Kling, and Jeffrey B. Liebman, "Moving to Opportunity in Boston: Early Results from a Randomized Mobility Experiment," *Quarterly Journal of Economics* 116 (May 2001): 607–54. The one random assignment experiment on teacher quality that we are aware of finds that Teach for America participants are somewhat more effective in teaching math (but not reading) to low-income children than the other teachers serving in these schools. See Paul Decker, Daniel Mayer, and Steven Glazerman, *The Effects of Teach for America on Students: Evidence from a National Evaluation* (Princeton, NJ: Mathematica Policy Research, 2004).

reaches 30 students, a school can apply to the ministry of education for another teacher, such that in the following year there will be an additional classroom and both will have fewer students.¹⁶ The characteristics of students in class sizes of 30 should be equal on average to the characteristics of students after the new classrooms are added, with the only difference being the number of students in the classroom. Consequently, the difference in student achievement before and after the application of the class size rule should provide an unbiased estimate of the class size effect. Using a regression-discontinuity research design, Urquiola finds that the mathematics achievement of third-grade students in a class of 20 students is 0.24 to 0.45 standard deviations (SD) higher than the achievement of students in a class of 30 students.¹⁷ Using a similar analytic strategy in Israel, Joshua Angrist and Victor Lavy find a comparable class size effect of 0.46 SD for fifth-grade mathematics.¹⁸ A remaining potential source of bias in these estimates, however, is that teachers hired to accommodate extra classrooms may be less effective. In this instance, the difference in achievement between the pre- and postrule group of students may be biased downward by the lower average quality of teachers in the latter group. In other words, this design cannot control for the effect of unobserved teacher characteristics.

Another recent approach to identifying class size effects has been to use panel data sets containing measures of student achievement at several points in time and information on teachers and their classes in several school years. Such data permit estimation of class size effects by comparing the performances of students taught by the same teacher in different school years. Studies using this methodology find smaller class size effects than those reported in studies using the regression discontinuity methodology.¹⁹ One possible explanation concerns differences in setting. Panel studies of class size effects utilize data sets from the United States, while studies using the regression discontinuity design utilize data from Bolivia and Israel.

¹⁶ Miguel Urquiola, "Identifying Class Size Effects in Developing Countries: Evidence from Rural Schools in Bolivia," *Review of Economics and Statistics* 88 (February 2006): 171–77.

¹⁷ Ibid.

¹⁸ Joshua D. Angrist and Victor Lavy, "Using Maimonides' Rule to Estimate the Effect of Class Size on Scholastic Achievement," *Quarterly Journal of Economics* 114 (May 1999): 533–75. Such policy rules have also been used in research from the United States, but contrary to research from developing countries, the effect of class size is not statistically significant. See Caroline M. Hoxby, "The Effects of Class Size on Student Achievement: New Evidence from Population Variation," *Quarterly Journal of Economics* 115 (November 2000): 1239–85.

¹⁹ By including dummy variables for every student and every teacher in the sample, studies using such data sets are able to eliminate all unobserved time-invariant student characteristics and teacher characteristics as sources of bias. Eric A. Hanushek, John F. Kain, and Stephen G. Rivkin ("Teachers, Schools, and Academic Achievement," NBER Working Paper no. 6691 [National Bureau of Economic Research, Cambridge, MA, 1998]) find that a 10-student difference in class size is associated with only a 0.04 to 0.06 SD difference in student achievement. Using panel data from New Jersey, Jonah E. Rockoff ("The Impact of Individual Teachers on Student Achievement: Evidence from Panel Data," *American Economic Review* 94 [May 2004]: 247–52) finds that the effect of class size on mathematics achievement is not statistically significant.

Researchers have also used panel data on teachers and students to examine the impact of teaching experience on teaching performance. One striking finding from these studies is that a teacher's effectiveness increases markedly with experience during the first few years on the job.²⁰

To our knowledge only one study of peer group composition effects among elementary or secondary school students makes use of data from a random assignment experiment. In this study, Bryan Graham, using data from the Tennessee STAR experiment, reports that the predicted math achievement of a primary school student placed in a class in which the average academic ability of classmates was one standard deviation above the average value for classrooms in the sample is 0.15 SD higher than the predicted math achievement of the same student placed in a class with peers of average academic ability.²¹

To date, the best estimates of peer effects in the developing world have been obtained from regression models that use school fixed effects to account for unobserved differences among schools. In this approach the effect of peer group composition is estimated by comparing achievement differences among students in different classrooms within the same school. Applying this strategy to data from Chile, Patrick McEwan finds that eighth graders who are in a classroom in which the average level of mother's education of classmates is one standard deviation above the average for classrooms in the sample score 0.287 SD higher on a test of mathematics achievement than do eighth graders who are in a class in which the average level of peers' mother's education is at the sample average.²² This estimate may still be biased by unobserved teacher characteristics, however, unless students are randomly allocated to classrooms within schools, which does not seem to be true for the Chilean data set.²³

Given the slow progress in understanding the roles that different resources play in determining why some schools are more effective than others, it is not surprising that progress has also been slow in understanding the relative effectiveness of public schools and private schools and the differences

²⁰ See Hanushek et al., "Teachers, Schools, and Academic Achievement"; and Rockoff, "Impact of Individual Teachers."

²¹ Bryan Graham, "Using Conditional Excess Variance Contrasts to Identify Social Interactions: Theory and an Application on the Relationship between Peer Groups and Academic Achievement," unpublished manuscript (Department of Economics, Harvard University, Cambridge, MA, 2004).

²² Patrick J. McEwan, "Peer Effects on Student Achievement: Evidence from Chile," *Economics of Education Review* 22 (April 2003): 131–41.

²³ McEwan ("Peer Effects on Student Achievement") also compares the test scores of siblings and twins who are assigned to different classrooms, i.e., by using siblings/twins fixed effects, and finds that a one standard deviation difference in the mean mother's education of one's classroom peers is associated with a difference in mathematics achievement of 0.371 and 0.536 SD for the sample of siblings and twins, respectively. These estimates will be biased, however, if siblings differ in ability and/or if students are nonrandomly assigned to classrooms.

between them.²⁴ In most studies, the private school effect is defined as the difference in achievement between public and private schools, net of the preexisting characteristics of their students. This effect is estimated by regressing achievement on a private school indicator and an array of student background variables. Research from Colombia using this methodology finds a private school advantage in the range 0.20 to 0.29 SD for secondary school achievement.²⁵ However, these studies fail to control for important student and school characteristics such as parents' expectations and school admissions policies, which may result in selection bias in ordinary least squares (OLS) estimates. Attempts have been made to correct this bias using standard Heckman selectivity bias corrections, but their application fails to meet critical assumptions.²⁶

More reliable estimates of the relative effectiveness of private schools come from randomized voucher experiments in which a "treatment" group of students is given vouchers to attend a private school and their post-treatment achievement is compared to that of students who were included in the lottery but did not receive vouchers. Most relevant to this article is a random assignment experiment conducted for secondary school students in Colombia that was evaluated by Joshua Angrist et al.²⁷ These authors report a private school advantage of 0.29 SD.

Although estimates from randomized experiments are clearly informative, they suffer from the limitation of treating schools as black boxes. As a result, they do not shed light on the reasons that private schools are more effective than public schools. This may matter, especially if peer group composition is an important determinant of student achievement. A combination of self-selection by parents and the operation of private school admission policies may create academically stronger peer groups in private schools that

²⁴ For reviews of the evidence, see Patrick J. McEwan, "The Potential Impact of Large-Scale Voucher Programs," *Review of Educational Research* 70 (Summer 2000): 103–49; and Richard J. Murnane, "The Role of Markets in American K–12 Education," in *The Limits of Market Organization*, ed. Richard R. Nelson, 161–84 (New York: Russell Sage, 2005). For evidence pertaining to Latin America, see Marie-Andrée Somers, Patrick J. McEwan, and J. Douglas Willms, "How Effective Are Private Schools in Latin America?" *Comparative Education Review* 48 (February 2004): 48–69.

²⁵ George Psacharopoulos, "Public vs. Private Schools in Developing Countries: Evidence from Colombia and Tanzania," *International Journal of Educational Development* 7 (January 1987): 59–67; Donald Cox and Emmanuel Jimenez, "The Relative Effectiveness of Private and Public Schools: Evidence from Two Developing Countries," *Journal of Development Economics* 34 (November 1991): 99–121.

²⁶ See Somers et al., "How Effective Are Private Schools," for further discussion.

²⁷ Joshua Angrist, Eric Bettinger, Erik Bloom, Elizabeth King, and Michael Kremer, "Vouchers for Private Schooling in Colombia: Evidence from a Randomized Natural Experiment," *American Economic Review* 92 (December 2002): 1535–58. For experiments in the United States, see W. G. Howell, P. J. Wolf, D. E. Campbell, and P. E. Peterson, "School Vouchers and Academic Performance: Results from Three Randomized Field Trials," *Journal of Policy Analysis and Management* 21 (Spring 2002): 91–217; Cecilia E. Rouse, "Private School Vouchers and Student Achievement: An Evaluation of the Milwaukee Parental Choice Program," *Quarterly Journal of Economics* 113 (May 1998): 553–602; and Eric Bettinger, "The Effects of Educational Vouchers on Academic and Non-academic Outcomes: Using Experimental Economic Methods to Study a Randomized Natural Experiment," unpublished manuscript (Weatherhead School of Management, Case Western Reserve University, 2003).

enhance their effectiveness. However, if this is an important part of the explanation for the relative effectiveness of private schools, public policies designed to increase student enrollments in private schools may result in a reduction in their effectiveness. Marie-Andrée Somers, Patrick J. McEwan, and J. Douglas Willms present evidence indicating that the private school effect for third- and fourth-grade language and mathematics achievement is no longer statistically significant after controlling for peer group SES.²⁸ In Colombia, more notably, the estimated private school effect is 0.32 SD but is reduced to 0.02 SD when controls for peer group SES are included in the model. Unfortunately, estimated effects from this study and others are likely to be biased by the omission of important student and school characteristics—hence the need for new research.

One other limitation of the literature on private school effects is that it does not explore the extent to which there is variation in quality among schools within each sector. This is critical because the greater the variation in quality among schools in a sector relative to the difference in average quality between sectors, the less important the decision about which sector in which to enroll a child and the more important the decision about which school within a sector in which to enroll a child. We now describe our approach to investigating these effects, which is based on a special property of our data set that allows us to address some of the limitations of prior work.

IV. Research Design

Data Set

To address our research questions, we designed a two-wave study of fifth-grade classrooms in public and private schools in Bogotá, Colombia. Beginning with the population of schools serving fifth-grade students in the city, we stratified schools by sector (public and private) and by school SES.²⁹ We then selected schools from these strata randomly (fig. A1 in the appendix contains a flowchart that describes how we arrived at the final analytic sample). In 1999, out of the 102 schools selected for the study, 14 refused to participate. One year later, we contacted the 88 remaining schools for follow-up. On this occasion, 10 schools refused to participate, one school did not offer fifth grade, and in four schools the fifth-grade mathematics teacher did not answer the questionnaire. Of the 3,095 fifth-grade students tested in 2000, 714 were either “new” to the school or had been absent from school the previous year on the day the fourth-grade tests were administered. For

²⁸ Somers et al., “How Effective Are Private Schools.”

²⁹ For administrative purposes, Bogotá is stratified into six distinct socioeconomic strata with six being the highest (see n. 10). Schools in Bogotá are highly segregated by SES.

these students, we imputed a pretest score using a hot-decking procedure.³⁰ The final sample includes 3,095 students from 73 schools, grouped in 97 fifth-grade classrooms and taught by 77 mathematics teachers. Of these students, 1,660 (54 percent) were enrolled in private schools and 1,435 students (46 percent) in public schools. Our sample includes 37 for-profit schools, nine religious schools, and one cooperative school.

A particular characteristic of this data set, which drives an important part of our analysis, is that, of the 77 teachers in the sample, 13 taught fifth-grade mathematics to multiple groups of children. As we discuss below, this characteristic of the data allows us to estimate the impacts of class size and peer group composition on student achievement, controlling for unobserved teacher characteristics.

Measures

Our data analyses involve three types of variables: outcome variable, question predictors, and control predictors. We measure these variables at four different levels: the student, the classroom, the teacher, and the school (table A1 in the appendix lists the variables and provides brief descriptions).

Student-Level Outcome and Control Predictors

Table 1 presents the means and standard deviations of the variables included in our data analysis for the full sample of students and separately for students in public schools and private schools. We discuss differences between the sectors later in this section.

Outcome variable.—Our outcome variable is each student's score (MSCORE) on an IRT-scaled mathematics test developed by UNESCO and the *Oficina Regional de Educación de la UNESCO para América Latina y el Caribe* (OREALC) and administered to students in the final weeks of their fifth-

³⁰ The personal and family characteristics used in this procedure included student's age, gender, number of days absent from school, mother's education, mother head of household, family's ownership of an encyclopedia, computer, car, refrigerator, washing machine, vacuum cleaner, phone, and having access to electricity in the home. All of the latter have been used as indicators of the student's SES. We also fitted our final models on the smaller sample of students for whom we had complete data. The results were not substantively different from those reported in the paper. In order to evaluate the generalizability of our study's findings to the population from which our sample was drawn, we assessed whether the schools and students that did not respond in the second wave of data collection differed in any systematic manner from those that did. On average, nonmatched students are more likely to be male, to have obtained lower scores on the tests, to come from single mother households, and to have been absent from school more often. Although proportionally more public schools dropped out of the study than did private schools, we found no statistically significant differences in school average pretest scores or school SES between public schools that participated both years and those that dropped out of the study after the first year. However, among private schools, those schools that participated in both years had, on average, higher pretest scores and student SES than those schools that dropped out after the first year.

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TABLE 1
MEAN (Standard Deviation) OF OUTCOME AND PREDICTORS, BY SECTOR

Variable	Full Sample*	Public†	Private‡
Outcome variable:			
MSCORE	297.6 (39.1)	292.3 (38.1)	302.2 (39.4)
Student-level control predictors:			
PRETEST	200.2 (50.1)	191.2 (49.4)	207.9 (49.4)
FEMALE	.57	.59	.56
AGE	10.8 (.97)	11.0 (1.1)	10.7 (.83)
DAYABS	.65 (1.4)	.73 (1.5)	.58 (1.3)
MHH	.25	.27	.22
MOTHED1	.06	.09	.04
MOTHED2	.26	.32	.20
MOTHED3	.37	.40	.35
MOTHED4	.31	.19	.41
BOOKS	0 (1.00)	-.16 (1.16)	.14 (.81)
HWHELP1	.07	.07	.07
HWHELP2	.81	.8	.82
HWHELP3	.12	.13	.11
Class-level predictors:			
CLMOTHED	2.9 (.48)	2.7 (.37)	3.1 (.47)
CLASS SIZE	36.8 (8.5)	40.2 (3.8)	33.9 (10.1)
Teacher predictors:			
EDLEVEL1	.05	.02	.07
EDLEVEL2	.01	0	.02
EDLEVEL3	.49	.30	.65
EDLEVEL4	.38	.68	.12
EDLEVEL5	.07	0	.13
TEAEXP	16.2 (9.8)	19.9 (7.7)	12.9 (10.1)

* *n* students = 3,095, *n* classrooms = 97, *n* teachers = 77, *n* schools = 73.† *n* students = 1,435, *n* classrooms = 38, *n* teachers = 29, *n* schools = 26.‡ *n* students = 1,660, *n* classrooms = 59, *n* teachers = 48, *n* schools = 47.

grade school year.³¹ The average mathematics score across all students in the sample was almost 300 points, with a standard deviation of about 40.

Control predictors.—We included as control predictors the average of each

³¹ This test was developed in 1997 for an international comparative study of third and fourth graders organized by UNESCO's Laboratorio Latinoamericano de la Calidad de la Educación. Thirteen Latin American countries, including Colombia, participated in this study. Curriculum and testing experts from all participating countries, with the help of the Educational Testing Service, developed and pilot tested the instruments to make them reflect the curriculum of the first 4 years of primary education in the participating countries. This test has two parallel forms (form A and form B), each containing 32 questions. We used both forms of this test in the study, randomly distributing each form to approximately half the students in each classroom. We included in all models a dichotomous predictor to distinguish the form of the fifth grade test taken by each student. The reported Cronbach's alpha reliability for forms A and B of the test is .86 and .88, respectively, as reported in UNESCO, *Primer estudio internacional comparativo sobre lenguaje, matemática y factores asociados en tercero y cuarto grado: Informe preparado por el Laboratorio de Evaluación de la Calidad de la Educación* (Santiago: UNESCO/OREALC, 1998), 67.

student's scores on achievement tests in mathematics and language taken at the end of the fourth grade (PRETEST) and its square.³² In doing this, we assume that the pretest scores capture students' total educational experience up to that point in time, as well as the effects of other earlier student background factors that may be influencing their achievement. Note that the pretest is scaled differently than the posttest, and as a result we were unable to use a change score as our outcome variable.

We included two student demographic characteristics as control predictors: (a) AGE, the student's chronological age, measured in years, and (b) FEMALE, student gender, measured as a dichotomous variable (coded one for female, and zero for male). Notice, in table 1, that, of the 3,095 students in the sample, 57 percent are girls. This is due to the inclusion of 19 all-female classes (and two all-male classes) in the sample of 97 classes. Also, boys were more likely to change schools and to be absent from school, preventing us from matching them on the 2 years of this study. Students in the sample are on average 10.8 years old, and their ages range from 8 to 17.

We used two sets of predictors as indicators of student SES: (a) MHH, a dichotomous indicator of whether the mother was head of the household (coded 0 = no, 1 = yes), and (b) indicators of the mother's educational attainment (MOTHED).³³ Approximately one-quarter of the students in the sample lived in mother-headed households. On average (median and mode), mothers had completed some high school.

We also included as control predictors measures of the student's school attendance, the availability of books in the home, and the support provided at home for the completion of homework, which previous studies in Bogotá had found to be related to student achievement.³⁴ We measured these variables as follows:

³² The fourth-grade tests used in this study were developed by Universidad Nacional for Bogotá's Department of Education 1998 student assessment program, where they were used to test third graders. The pretests had three parallel forms, each containing 17 questions on mathematics and 17 on language. We administered all three forms of the pretest, randomly distributing the forms to approximately one-third of the students in each classroom. Items were scored separately by subject, on a scale ranging from 0 to 306. Students achieved an average of 200.2 points (SD = 50) with scores ranging from 12 to 306 points (six students in the sample obtained perfect scores). There are no published reliability or validity estimates for the pretests. We verified that the substantive results were not different when the models were fit with both the fourth-grade reading score and math score included separately as control variables.

³³ We chose to use mother's education rather than father's education for two reasons. First, there are a number of studies that suggest that mother's education is more important than father's education in determining IQ; see Arleen Leibowitz, "Home Investments in Children," *Journal of Political Economy* 82 (March 1974): S111-S131. Second, there was a high percentage of missing data on father's education in our sample. In this sample, 356 observations (9.9 percent of the student sample) were missing information on mother's education. We imputed these missing values by substituting the median value for the student's classmates (we imputed the median rather than the mean because we did not want the imputed values to be sensitive to outliers). Likewise, 178 observations (4.9 percent of the student sample) were missing data on whether their mother was head of the household (MHH). Given that we constructed this variable based on students' responses to whether their father lived with them or not, in the cases where they failed to answer we assumed that their father did not.

³⁴ Sarmiento et al., *Evaluación de la Calidad*.

DAYABS, the numbers of days that the student was absent from school over the previous 2 weeks;³⁵

HELPHW, the frequency with which a student receives help with his or homework at home, measured on a Likert-type scale (where 1 = never receives help, 2 = sometimes receives help, and 3 = always receives help);³⁶ and

BOOKS, measuring the availability of books in the student's home, a composite variable standardized to a mean of zero and standard deviation of one.³⁷

In spite of the low levels of mother's education among students in the sample, 97 percent of students reported having books in their homes, and 93 percent reported receiving help sometimes or always with their homework at home. Students in the sample were absent from school an average of 0.65 days in the 2 weeks preceding completion of the questionnaire.

Question Predictors

To answer our research questions on the influence of classrooms, teachers, and peers on student achievement, we included in the data analyses four question predictors. Two of these predictors are at the classroom level, and two are at the teacher level.

Classroom characteristics.—The predictor that we used to represent peer group composition is the “leave-out” mean of the student's classmates' mothers' educational level (CLMOTHED).³⁸ This is defined as the classroom mean of mother's education, leaving out each student's own value from the calculation. As can be seen in table 1, students in the sample are likely to have a peer group whose average level of mother's education is an incomplete secondary school education, but there is considerable variation across classrooms.

We measure class size as the number of students enrolled in each classroom as reported by the teacher. On average, classrooms in the sample have 37 students (SD = 8.5) although the range extends from 10 to 53 students.³⁹ Based on exploratory analyses that revealed a nonlinear relationship between class size and student achievement, we defined our class size predictor (CLAS-

³⁵ In the 77 cases (2.8 percent of the student sample) in which information was missing on this variable, we substituted the sample mean for the missing value.

³⁶ In the data analyses, we converted this ordinal variable into a system of dichotomous (dummy) predictors, with HWHLP3 (always receives help) being omitted from the fitted models as a reference category. In the 19 cases (0.7 percent of the student sample) in which information was missing for this variable, we substituted the value 2, which was the modal response.

³⁷ For the 15 cases (0.5 percent of the student sample) in which information was missing on this variable, we assumed that there were no books in the home.

³⁸ See Michael A. Boozer and Stephen E. Cacciola, “Inside the ‘Black Box’ of Project Star: Estimation of Peer Effects Using Experimental Data,” Economic Growth Center Discussion Paper no. 832 (Yale University, New Haven, CT, 2001).

³⁹ Averaged over students rather than classrooms, the mean is 36 students.

SIZE) as the natural logarithm of the sample maximum class size (53) minus the student's own class size, or $\ln(53 - \text{class size} + 0.167)$.⁴⁰

Teacher characteristics.—We included as predictors two sets of characteristics of teachers that are rewarded by the existing salary structure. The first measures educational preparation to teach. The original ordinal variable recorded the teacher's highest educational degree obtained; these were coded 1 = pedagogical high school; 2 = education technology (2- or 3-year post-secondary certificate); 3 = a 4-year university program in education; 4 = postgraduate courses, including master's in education (2 years plus thesis) or specialization (1- or 2-year graduate-level program); 5 = graduate from a 5-year university-level program in a field other than education. In our analyses, we converted this ordinal variable into a system of dichotomous (dummy) predictors, with completion of a 4-year university program in education (the modal value in our analytic sample) omitted from the fitted models as a reference category.

We measured teacher experience (TEAEXP) by the number of years that the fifth-grade mathematics teacher had been in the teaching profession. We included the natural logarithmic transformation of this variable (L_TEAEXP) as a predictor in the regression analyses, given that previous research suggests that the relationship between student achievement and teachers' years of experience is nonlinear.⁴¹ On average, teachers in the sample have 16 years of teaching experience (SD = 9.8), while 11 percent have 3 years or less of experience.

School sector.—In order to examine the effect of school sector on student achievement, we included a dichotomous predictor (PRIVATE), coded private = 1 and public = 0. Fifty-four percent of the students in the sample

⁴⁰ We chose a log function over a quadratic function of class size for several reasons. First, a quadratic functional form results in predicted values that first increase and then decrease very sharply with class size. Therefore, we chose a log functional form because it constrains achievement to be a monotonic function of class size. Second, we could not reject the null hypothesis that the predicted values of student achievement from models with the alternative functional forms for class size were not different, when evaluated at either the tenth or ninetieth percentile of the class size distribution. Third, the model's fit was slightly better with the log specification than with the quadratic specification. Note that we followed Frederick Mosteller and John W. Tukey's advice and "started" the transformation by adding a small amount (0.167) to the difference in order to avoid infinities when the student's class size equaled the maximum; see their *Data Analysis and Regression* (Boston: Addison-Wesley, 1977), chap. 4. To verify the appropriateness of this recommendation, we estimated a nonlinear model of student achievement against $\ln(\text{Max Class Size} + k - \text{Class Size})$ in order to determine the value of k that maximized the model fit. This procedure did indeed return a value very close to the recommended value.

⁴¹ Hanushek et al., "Teachers, Schools, and Academic Achievement"; Richard J. Murnane and Barbara R. Phillips, "Learning by Doing, Vintage, and Selection: Three Pieces of the Puzzle of Relating Teaching Experience and Teaching Performance," *Economics of Education Review* 1 (Autumn 1981): 453–65; Rockoff, "Impact of Individual Teachers." As in the case of class size, we chose the log functional form over the quadratic functional form because the pattern of predicted values resulting from the latter seemed unlikely, given previous research (such as that by Hanushek et al.). The difference in predicted values between the log and quadratic functional forms is not statistically significant at either the tenth or ninetieth percentile of experience. Moreover, model fit is virtually identical for the two models.

attended private schools. A total of 59 classrooms located in 47 private schools are included in the sample. Thirty-seven of the private schools in the sample are for-profit schools, nine are run by religious organizations, and one is a cooperative school.

Data-Analytic Strategy

The goal of our analysis is to investigate to what extent differences in class size, peer group composition, and teacher quality account for differences between the effectiveness of public and private elementary schools in Bogotá. Before proceeding to a comparison of achievement across sectors, however, we address two baseline questions: (a) whether student achievement varies across classrooms, net of student characteristics and prior achievement, and (b) whether achievement, net of background characteristics, varies among students in different classrooms taught by the same teacher.⁴² Next we investigate whether class size and peer characteristics explain differences in achievement among classrooms, irrespective of sector. Finally, we estimate the extent to which class size, peer group composition, and teacher quality explain differences in achievement between the public and private sector.

In these analyses, to ensure that standard errors are estimated correctly, we fit all regression models using either fixed effects or random effects to account for the clustering of students within classrooms (see our Analytic Appendix for a definition and discussion of these two terms and an explanation of how they relate to the multilevel modeling strategy of Raudenbush and Bryk).⁴³ The following section provides greater detail on each of the statistical models fitted in our analysis.

Research Questions 1a and 1b

- 1a. Do children enrolled in some fifth-grade classrooms in Bogotá have higher end-of-year mathematics achievement, on average, than children in other classrooms, controlling for student SES and end of fourth-grade achievement?
- 1b. Do children enrolled in some fifth-grade Bogotá classrooms have higher end-of-year mathematical achievement, on average, than children enrolled in other fifth-grade classrooms taught by the same teacher?

We address these baseline questions using fixed effects regression analysis in which we model the relationship between a student's fifth-grade mathematics achievement score and a set of dummy predictors representing classroom

⁴² If student achievement, net of background characteristics, did not vary among classrooms, our research could not provide any results. However, as explained below, student achievement is higher in some classrooms than in others, and this is even true among classrooms taught by the same teacher.

⁴³ Stephen W. Raudenbush and Anthony S. Bryk, *Hierarchical Linear Models: Applications and Data Analysis Methods* (Thousand Oaks, CA: Sage, 2002).

membership, controlling for a vector of student background characteristics

$$Y_{ij} = \sum_j \beta_{0j} D_j + \beta_1 S_{ij} + \varepsilon_{ij}, \quad (1)$$

where variables and parameters are defined as follows:

- Y_{ij} = fifth-grade mathematics achievement of student i in classroom j ;
- D_j = vector of classroom dummy predictors (one per classroom);
- S_{ij} = vector of grand mean-centered individual-level student characteristics for student i in classroom j ;
- β_{0j} = intercept ("fixed effect") for classroom j (the average mathematics achievement of students in classroom j , adjusted for the included student control characteristics);
- β_1 = parameter vector representing the effects of the student characteristics;
- ε_{ij} = residual for student i in classroom j .

To answer question 1a, we test the null hypothesis that all classroom fixed effects (intercept parameters β_{0j}) in model 1 are simultaneously equal. If we reject this null hypothesis, we can conclude that the fifth-grade classroom to which the student was assigned influenced his or her end of fifth-grade mathematics achievement score, net of included student control characteristics. To answer question 1b, we test the null hypothesis that all classroom intercept parameters β_{0j} in model 1 that pertain to classrooms taught by the same teacher are simultaneously equal. If we reject the null hypothesis, we can conclude that average student achievement differs among classrooms taught by the same teacher, net of included student control characteristics.

While the above strategy can detect the presence of classroom or teacher effects, it does not reveal which specific classroom and teacher characteristics are responsible for the effects. Our next research questions examine this issue.

Research Question 2

2. Do the student body composition and number of students in the class in which a student spent the fifth grade influence his or her fifth-grade mathematics achievement score?

To answer this question, we fitted a set of regression models in which we regress a student's mathematics achievement score against the characteristics of his/her classroom, while also including a set of dummy predictors representing the teacher to which they are assigned (rather than the classroom to which they are assigned, as in model 1) and controlling for student background characteristics. With this strategy, we examine whether differences in class size and peer group composition among classes taught by the same teacher explain differences in student achievement. The regression model

has the following form:

$$Y_{ijk} = \sum_K \beta_{0k} D_k + \beta_1 S_{ijk} + \beta_2 C_{jk} + \varepsilon_{ijk}, \quad (2)$$

where additional variables and parameters are defined as follows:

- Y_{ijk} = fifth-grade mathematics achievement of student i in classroom j with teacher k ;
- D_k = vector of teacher dummy predictors (one per teacher);
- S_{ijk} = vector of grand mean-centered individual-level student characteristics for student i in classroom j with teacher k ;
- C_{jk} = vector of grand mean-centered classroom and peer group characteristics of class j with teacher k ;
- β_1 = parameter vector representing the effects of the student characteristics;
- β_2 = parameter vector representing the effects of the classroom and peer group characteristics;
- ε_{ijk} = residual for student i in classroom j with teacher k .

Our strategy of including a set of dummy predictors to represent all observed and unobserved differences in the outcome among teachers (the fixed effects of teachers) resolves a problem that has plagued previous studies, namely, the nonrandom assignment of students to teachers. Once teacher fixed effects are included in the model, we are still able to estimate the impact of the class size and classroom-level peer group composition on student achievement because our data set contains several teachers who teach in multiple classes, thereby providing classroom-level outcome variation that is not eliminated by the inclusion of the teacher fixed effects.

However, our strategy will produce unbiased estimates of class size and peer group effects (β_2) only if students are randomly assigned to different classes taught by the same teacher. To learn about assignment processes, we interviewed a teacher or administrator in each of the 13 schools in which individual teachers taught mathematics to more than one fifth-grade class. Our respondents reported that, in all but two cases, students were assigned to classes to obtain a mix of student backgrounds and abilities in each class and to roughly equalize class sizes. In two cases students with especially low or high skills were assigned to a particular class. We eliminated these classrooms from the analytic sample.⁴⁴

⁴⁴ To verify our respondents' statements that, with the exceptions noted above, students taught by the same teacher were not assigned to classes on the basis of ability or SES, we utilized a strategy proposed by McEwan ("Peer Effects on Student Achievement"). This involved creating "virtual classrooms" by randomly reassigning students from separate classrooms into "new" classrooms. We then compared the distribution of class size and peer group composition in the virtual classrooms with those in the actual classrooms. We found almost no differences in the two distributions. Of course, this means that there is only modest variation in class size and peer group composition among classrooms taught by the same teacher.

In summary, we believe that our method of making use of a unique property of our Bogotá data set provides new insights into the roles that class size and peer group composition play in explaining the relative effectiveness of public and private schools. Nevertheless, our empirical results are by no means definitive, because they are based on estimates from student achievement in the modest number of classrooms (31) taught by the 13 teachers who taught more than one fifth-grade class.

Research Questions 3a, 3b, and 3c

- 3a. Controlling for student background characteristics, do students enrolled in private schools have higher achievement than students enrolled in public schools?
- 3b. Controlling for student background characteristic and peer group composition, do students enrolled in private schools have higher achievement than students enrolled in public schools?
- 3c. Controlling for student background characteristics, peer group composition, and class size, do students enrolled in private schools have higher achievement than students enrolled in public schools?

To address question 3a we fit the random effects regression model below (model 3). This model is similar to our model 1, except that the classroom-specific intercepts (the classroom fixed effects) have been converted into the corresponding random effects, u_j , in order that school sector (PRIVATE) can be included as a predictor:⁴⁵

$$Y_{ijk} = \beta_0 + \beta_1 S_{ijk} + \beta_3 \text{PRIVATE}_{jk} + u_j + \varepsilon_{ijk}, \quad (3)$$

where

- β_0 = intercept parameter representing the average fifth-grade student mathematics achievement (across all classrooms and teachers), controlling for all predictors in the model;
- β_3 = effect of the school sector of classroom j with teacher k on fifth-grade student mathematics achievement, controlling for all other predictors in the model;
- u_j = random effect of classroom j on fifth-grade student mathematics achievement.

To address question 3b, we add the predictor representing peer group composition to the model described above (to provide model 3b). To address question 3c, we further add the class size predictor (to provide model 3c).

Unlike the corresponding estimates obtained from our teacher fixed effects models (model 2), it is possible that the estimated effect of school sector obtained in our classroom random effects models (model 3) may be

⁴⁵ This random effects specification is essentially a special case of the most general specification used by Raudenbush and Bryk, *Hierarchical Linear Models*; see the Analytic Appendix for details.

biased by the omission of unobserved student and classroom characteristics. However, we can reduce the magnitude of the bias in the school sector effect in models 3b and 3c by constraining the effects of the class size and peer group composition to the estimates that we obtained in our teacher fixed effects specification (model 2) and by controlling for students' pretest scores. The logic of constraining these coefficients in this way is that, by virtue of its fixed effects specification, model 2 takes into account all unobserved teacher quality differences with which class size and peer attributes may be correlated.⁴⁶

Research Question 4

4. Does a student's math achievement at the end of fifth-grade depend on his or her fifth-grade math teacher's qualifications and years of teaching experience?

To address this research question, we supplement the hypothesized random effects model in model 3, above, as follows:

$$Y_{ijk} = \beta_0 + \beta_1 S_{ijk} + \beta_2 C_{jk} + \beta_3 \text{PRIVATE}_{jk} + \beta_4 \text{TEACHER}_{jk} + u_j + \varepsilon_{ijk}, \quad (4)$$

where

β_4 = effect of a vector of teacher qualifications and teaching experience (TEACHER_{jk}), controlling for vectors of student and classroom characteristics.

Again, and for the same reasons, we constrain the coefficients on the class size and peer group composition measures to their values estimated from fitting model 2.

V. Findings

Do Classrooms Matter to Student Mathematics Achievement?

The fifth-grade classroom to which a student was assigned makes a difference to his or her end of fifth-grade mathematics achievement score. Based on the results of a test of the equality of the classroom fixed effects across the 97 classrooms (obtained from model 1), we can reject the null hypothesis that there are no differences among classrooms ($p < .001$).⁴⁷ Accounting for imprecision in the estimates, the interquartile range of the estimated true classroom fixed effects is 9.8 points. This is about half of the difference between the median math achievement scores of third graders and fourth graders when the same test was administered to large samples of students in

⁴⁶ We also tested whether including interactions between PRIVATE and the other predictors in the model substantially improved model fit. We could not reject the null hypothesis that the interaction terms are jointly equal to zero ($\chi^2 = 11.91$, $p = 0.4528$, $df = 12$, based on model 3b). We therefore include only the main effect of PRIVATE in our reported analyses.

⁴⁷ $F(96, 2986) = 4.30$; $p < 0.001$.

these grades in Colombia in 1999.⁴⁸ We also find that there are statistically significant differences in math achievement, net of family background, for students enrolled in different classes taught by the same teacher. This is important because it means that it is possible, in subsequent analyses, for class size and peer group composition to make a difference in explaining why student achievement is higher in some classrooms than in others taught by the same teacher.

Before turning to the roles of specific classroom attributes in predicting student math achievement, we discuss briefly those student-level characteristics that have the strongest impacts on student achievement within classrooms (see app. table A2, first column of estimates in top half). First, as many other studies have shown, a student's achievement score at the end of one grade (fourth) is a strong predictor of the student's achievement at the end of the next grade (fifth). Second, on average, boys outperform girls by an average of 8 points ($p < .001$). Third, younger students perform slightly better on average than older students ($p < .001$), a likely result of accumulated academic failure among the latter. Fourth, fifth-grade mathematics achievement also depends upon school attendance, with students who were absent one additional day in the 2 weeks before the math test was administered having scores that are 1.9 points less, on average ($p < .001$).

Do Peer Group Composition and Class Size Matter to Student Mathematics Achievement?

As other researchers have pointed out, the estimation of peer effects and class size effects has often been biased as a result of failing to account for the nonrandom assignment of students to teachers.⁴⁹ To minimize this problem, we rely on variation in peer group composition among classes taught by the same teacher in the same school.

Columns 2–4 of table 2 contain estimates of parameters from models 2a, 2b, and 2c, which include fixed effects for teachers and different combinations of the two classroom-specific predictors, class size and peer group composition. As indicated in models 2b and 2c in columns 3 and 4, the point estimates suggest that students in classrooms in which their peers have more educated mothers achieve higher mathematics scores. While the relevant coefficients in the teacher fixed effects models are not statistically significant, they are similar in value to the statistically significant coefficients in the analogous models fitted with random effects in columns 6 and 7.⁵⁰ Based on the

⁴⁸ UNESCO, *Primer estudio internacional*, 258.

⁴⁹ Caroline M. Hoxby, "Peer Effects in the Classroom: Learning from Gender and Race Variation," NBER Working Paper no. 7867 (National Bureau of Economic Research, Cambridge, MA, 2000); McEwan, "Peer Effects on Student Achievement"; Boozer and Cacciola, "Inside the 'Black Box.'"

⁵⁰ See the Analytic Appendix for a more detailed explanation of our complementary use of both fixed effects and random effects models (the latter being most similar to the hierarchical linear models described in Raudenbush and Bryk, *Hierarchical Linear Models*). In essence, the parameter estimates from fixed effects models are less biased than those from analogous random effects models. However, because there are only 13 teachers in our sample who teach multiple classes, the standard errors of the estimates

TABLE 2
FIXED AND RANDOM EFFECTS REGRESSION MODELS SHOWING THE ESTIMATED EFFECTS OF
PEER GROUP COMPOSITION AND CLASS SIZE ON STUDENTS' FIFTH-GRADE
MATHEMATICS ACHIEVEMENT SCORE

	Teacher Fixed Effects			Classroom Random Effects		
	2a	2b	2c	2a (r.e.)	2b (r.e.)	2c (r.e.)
Class-level predictors:						
CLMOTHEd		12.199	7.932		13.805***	12.698***
CLASSIZE	26.534*		23.919 ⁺	-.483		-1.011
Goodness of fit:						
Overall R^2	.351	.350	.351	.290	.312	.312

NOTE.—r.e. = random effects; n students = 3,095, n classrooms = 97, n teachers = 77. Class size is represented by a modified log transformation described in Sec. III of the text. Under this transformation, smaller values of class size correspond to larger values in the transformed predictor. Coefficients associated with the teacher fixed effects and the control predictors have been omitted from the table to save space. (The regression coefficients associated with the control predictors can be found in app. table A2.)

⁺ $p < .10$.

* $p < .05$.

*** $p < .001$.

fitted fixed effects from model 2b in column 3, a one SD difference in the average value of mothers' education is associated with a 0.150 SD difference in individual students' math achievement. This is a little more than half the size of the analogous peer group effect of 0.287 SD estimated by McEwan using data from Chile.⁵¹ The difference may stem from differences in the age of the students, or differences in model specification—our model controls for unobserved teacher characteristics.⁵² Our estimate of the magnitude of the peer effect is almost identical to that found by Graham in his analysis of the Tennessee STAR experiment.⁵³

As indicated in fitted fixed effects models 2a and 2c in columns 2 and 4, class size is a statistically significant predictor of student achievement. (Fig. A2 in the appendix presents this fitted relationship.) Based on this fitted model, students in a classroom with 30 students are predicted to score 0.35 SD higher (13.5 points) than students in a classroom with 40 students. The magnitude of the impact is quite similar to that reported by Urquiola in Bolivia and Angrist and Lavy in Israel (0.24–0.45 and 0.46 of a standard

from fixed effects models will be larger than those from random effects models, resulting in a loss of power when making statistical inferences. Fortunately, in the case of model 2, the estimates from the fixed effects and random effects models are very similar, so we can conclude that the random effects model provides unbiased estimates after all. As such, we can use the (more precise) standard errors from the random effects model to make statistical inferences.

⁵¹ McEwan, "Peer Effects on Student Achievement."

⁵² Given our identification strategy, it is possible that the peer effect actually reflects differences in the effectiveness with which the same teacher teaches groups of students with different characteristics.

⁵³ We also tried estimating peer effects using an alternate measure of peer composition: the percentage of students in classroom j with two parents. In the fixed effects specification (model 2b), a 1 percent difference in the number of children from a two-parent family results in a 35.3 point difference in a classroom's average test score (and is statistically significant at the 5 percent level). Thus using this alternate measure also leads us to conclude that peer effects are statistically significant in our data set.

deviation higher in mathematics achievement in a class that is 10-students smaller, respectively).

Notice in table 2 that, while class size is a strong predictor of student achievement in the models fitted with teacher fixed effects, this is not the case in the models fitted with classroom random effects. The explanation is that, in the Bogotá schools included in our sample, the most effective teachers were assigned to work with the largest classes. Consequently, we argue that estimates of the class size effect from the random effects models—that do not fully control for differences in teacher quality—are biased.

One implication of this pattern is that estimates of the relative effectiveness of individual teachers taken from models that control for class size and peer group effects are quite different from estimates taken from models that do not control for these factors. For example, one teacher in our sample ranked forty-third in terms of average student achievement when only the student level characteristics were controlled; this teacher ranked first when both peer group composition and class size were included as predictors. This finding raises serious questions about school accountability initiatives that attribute differences in average student achievement across classrooms solely to differences in teacher quality.

Does School Sector Matter to Student Achievement?

As shown in table 1, on average students in our Bogotá data set who attended private schools, compared to those students who attended public schools, are more economically advantaged, are less likely to live in mother-headed families, and have mothers with more education. The differences in individual student characteristics between public and private schools naturally result in corresponding differences in the characteristics of peer groups by sector. As a result, public school students have less advantaged peers than do private school students. Public school students are also likely to study in classrooms with larger class sizes than are private school students.

One respect in which public school students have a potential advantage over students in private schools is in the academic preparation and experience of their teachers. No public school teacher in the sample has less than 6 years of teaching experience in contrast to 28 percent of private school teachers. While almost 70 percent of public school teachers have a master's degree or are working toward one, the corresponding figure for private school teachers is 12 percent.⁵⁴

We turn now to the question of whether the differences in peer group composition, class size, and teacher characteristics contribute to differences

⁵⁴ As explained earlier, this pattern stems primarily from the public school teacher salary scale and its applicability to private schools. While the salary schedule provides strong incentives for public school teachers to gain educational credentials and to remain in teaching, the salary scale creates incentives for private schools to avoid hiring highly experienced teachers with significant educational credentials because they are very expensive.

PRIVATE SCHOOLING IN BOGOTÁ

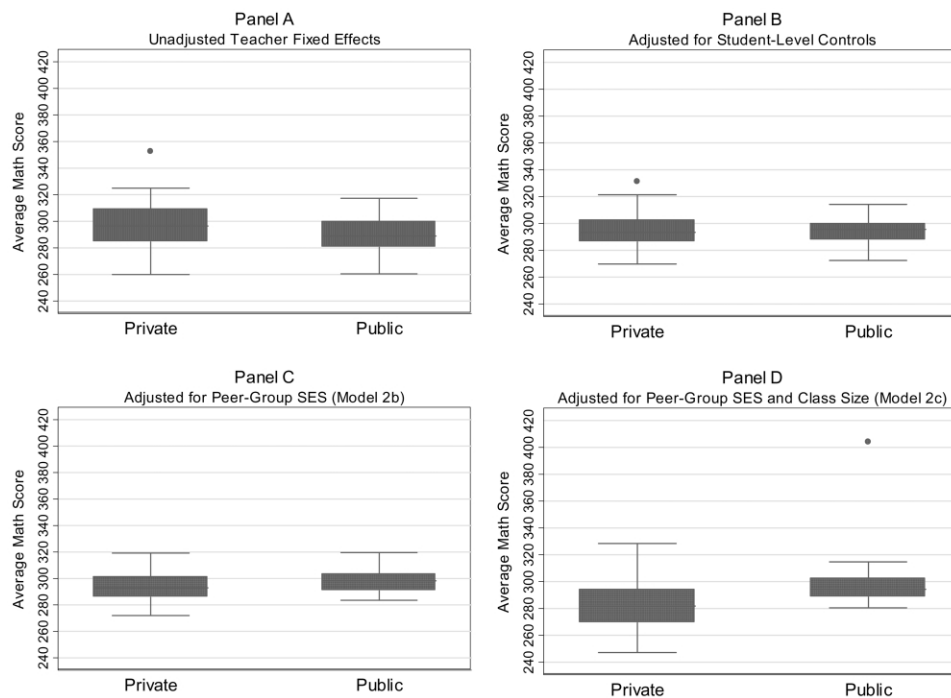


FIG. 1.—Distribution of estimated teacher fixed effects by sector and model specification. Panel B is based on fitted values from a version of model 2 that omits classroom characteristics, panel C is based on fitted model 2b, and panel D is based on fitted model 2c.

in the achievement of students attending schools in the two sectors. To find an answer, we look at the average achievement difference between public and private schools, both before and after controlling for differences in student, classroom, and teacher characteristics (as indicated in app. table A3 by the estimated coefficient of PRIVATE in the different specifications of model 3). For each school sector, we also examine the variability in average student achievement among groups of students taught by different teachers, both with and without controls for student and classroom characteristics. This variability is illustrated in figure 1 (details on the construction of this figure are provided in the following discussion).

Panel A of figure 1 displays the distribution of average teacher-specific fifth-grade student math scores among classrooms in public schools and private schools in the sample. The mean of the distribution for students in private schools is eight points (0.2 SD) higher than the mean for students of public schools (based on the estimated coefficient of PRIVATE in model 3a* of app. table A3, which does not control for student characteristics). This is not a surprise given the difference between the average SES of private school students and public school students. What is striking is the wide variation in teacher-specific average achievement among teachers in both sec-

tors, as seen in panel A of figure 1. This highlights that private schools in the sample do not all serve high achieving students.

Panel B of figure 1 displays the distribution of predicted average student achievement for different teachers, after controlling for student background characteristics, including prescore, by sector.⁵⁵ The difference in the adjusted average achievement of students in the two sectors is 0.8 points (0.02 SD) and not statistically significantly different from zero (see app. table A3, model 3a). This estimate is much smaller than the Colombian estimate from Somers, McEwan and Willms (0.32 SD), even though both estimates were obtained from a similar model and from data sets of students of comparable age.⁵⁶ It is also smaller than that reported by Angrist et al. in their evaluation of the voucher experiment in Colombia (0.29 SD).⁵⁷ A plausible explanation is that the mix of private schools in the country as a whole is different from the mix of private schools in our Bogotá sample.⁵⁸

Panel C of figure 1 displays the distribution of predicted average achievement of students taught by different teachers taken from fitted model 2b, which controls for peer group composition as well as for student background, by sector.⁵⁹ Net of the peer group effect, the adjusted achievement of students in private school classes is four points (or 0.10 SD) lower than that of students in the public sector (see app. table A3, model 3b). This highlights that a high-SES peer group is one of the relative advantages many private schools offer. However, again it is striking how great the variation in average student achievement is among students enrolled in the classes of different private school teachers.

Finally, panel D of figure 1 displays the distribution of predicted average achievement of students taught by different teachers taken from fitted model 2c, which controls for class size as well as for peer group and student background characteristics.⁶⁰ The adjusted mean achievement for private school students is now 14.5 points (or 0.36 SD) lower than the analogous mean for public school students (see app. table A3, model 3c). The explanation is that class size and peer group composition matter to student achievement and that, on average, class sizes are smaller and peer groups are more advantaged in the private sector.

⁵⁵ The data points used to construct panel B are the teacher fixed effects (β_{0k}) from the fitted model $Y_{ik} = \sum_k \beta_{0k} D_k + \beta_1 S_{ik} + \varepsilon_{ik}$, where the D_k are a set of dummy variables representing student i 's teacher, and S_{ik} is a vector of student i 's background characteristics. In other words, this is simply a version of model 2 that omits classroom characteristics. The estimated β_{0k} are too numerous to list in a table; however, they can be provided by the authors upon request.

⁵⁶ Somers et al., "How Effective Are Private Schools."

⁵⁷ Angrist et al., "Vouchers for Private Schooling."

⁵⁸ The statistically significant variation in predicted average student achievement among students taught by different private school teachers indicates how sensitive the size of the private school effect is to the composition of the private school sample.

⁵⁹ The data points used to construct panel C are the β_{0k} from fitted model 2b.

⁶⁰ The data points used to construct panel D are the β_{0k} from fitted model 2c.

Interpreting the estimated teacher fixed effects displayed in panel D as our best estimates of teacher quality, the wide range of quality estimates for private school teachers is remarkable, especially relative to the magnitude of the variation in estimated effectiveness among public school teachers. The best private school teachers appear to be more effective than the best public school teachers, while the worst private school teachers appear to be considerably less effective than the worst public school teachers.⁶¹

The panels of figure 1 shed light on differences in the ways that public schools and private schools “produce” student achievement. As indicated in panel B, on average, private schools in our Bogotá sample are as effective as public schools in producing student achievement. However, schools in the two sectors use very different mixes of resources. On average, private schools have teachers with fewer credentials and less experience than do public schools, but they compensate by having smaller classes and somewhat more advantaged student peers (panel D).

Do Teacher Qualifications and Experience Matter to Student Achievement?

The difference between the two sectors in the average quality of teachers and the extraordinary variation in teacher quality among teachers in the private sector prompts the question: do teaching experience and educational credentials, the attributes rewarded based on the national teachers’ contract, also predict teaching performance? To answer this question, we fitted the set of regression models presented in table 3 that estimate the effects of teacher education and years of experience on student mathematics achievement, with classrooms treated as a random effect. An important disadvantage of using a random effects specification, however, is that the estimated effects of class size and peer group composition will be biased, given that these variables may be correlated with unobserved teacher quality. In estimating models 4a and 4b, we therefore constrain the parameters associated with the peer group composition and class size predictors to their values estimated in the fixed effects analysis (model 2c), which are not biased by unobserved teacher quality (see the Analytic Appendix for a more detailed description of these issues). For comparative purposes, we display the results from fitting these models with the peer group composition and class size parameters left unconstrained (models 4c and 4d).

Figure 2 illustrates the fitted relationship between student achievement and teacher experience from model 4b (which does not control for sector). Notice that the slope of this relationship is steeper during a teacher’s first years in the classroom, after which the fitted relationship flattens out. This

⁶¹ It is important to point out that we do not know for sure that the average differences in achievement among students in classrooms taught by different teachers represent teacher effects. Although we do control for class size and one measure of peer group composition, it is possible that there are also unmeasured classroom-specific differences that account for some of the classroom-specific achievement differences.

TABLE 3
TAXONOMY OF FITTED CONSTRAINED AND UNCONSTRAINED RANDOM EFFECTS MODELS SHOWING THE
ESTIMATED EFFECTS OF TEACHER CHARACTERISTICS ON STUDENTS' FIFTH-GRADE MATHEMATICS
ACHIEVEMENT SCORES

	Classroom Random Effects (CLASSIZE and CLMOTHEd Constrained to the Values of Their Estimates in Fitted Model 2C)		Classroom Random Effects (Unconstrained Models)	
	4a	4b	4c	4d
Class-level predictors:				
CLMOTHEd	(7.932)	(7.932)	16.154***	15.694***
CLASSIZE	(23.919)	(23.919)	-.802	-1.033
School sector (PRIVATE = 1):				
PRIVATE	-12.695*		-1.603	
Teacher predictors:				
Years of experience:				
L_TEAXP	3.796	5.918*	.097	.327
Level of education (licensure or EDLEVEL3 omitted category):				
EDLEV1	-10.673	-11.842	-2.766	-2.840
EDLEV2	-23.592 ⁺	-27.053*	-2.345	-2.683
EDLEV4	-3.479	2.405	4.837 ⁺	5.438*
EDLEV5	5.107	3.233	-2.405	-2.838
Test of joint significance of EDLEV (χ^2)	5.55	7.05	4.70	7.98 ⁺
Goodness of fit:				
R ² "overall"	.251	.206	.318	.318

NOTE.— n students = 3,095, n classrooms = 97, and n teachers = 77. Estimated regression parameters associated with the control predictors can be found in app. table A2.

⁺ $p < .10$.

* $p < .05$.

*** $p < .001$.

supports the findings from other recent studies that suggest that the greatest gains from additional experience occur during the first years of teaching.⁶² Notice, however, that in models 4c and 4d the coefficient on this predictor is close to zero, suggesting that more experienced teachers appear to be assigned to classes that are larger and populated by more needy groups of students, thus exerting a compensatory effect that neutralizes the effects of experience.

Notice that in models 4a and 4b teachers who have less than a 4-year college education are less effective, on average, than those with a 4-year degree. Nine percent of the teachers in private schools in the sample have less than a 4-year degree compared to 2 percent of public school teachers. Thus, differences in educational credentials and years of teaching experience help to explain the difference between the average quality of private school teachers and those of public school teachers.

⁶² Other studies that show this same pattern include Hanushek et al., "Teachers, Schools, and Academic Achievement"; Murnane and Phillips, "Learning by Doing"; and Rockoff, "Impact of Individual Teachers."

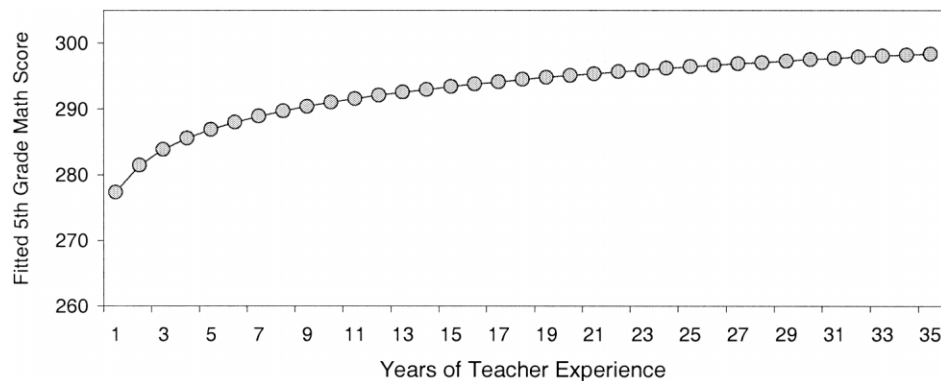


FIG. 2.—Fitted relationship between fifth-grade mathematics achievement score and years of teacher experience for a prototypical student in a classroom of average size and average peer group SES (from model 4b; n students = 3,095, n classrooms = 97, n teachers = 77).

It is also important to point out that the estimates for models 4a and 4b indicate that teachers with educational credentials beyond a 4-year degree (EDLEVEL4) are not more effective, on average, than are those with a 4-year degree. This supports the view of many Bogotá school officials that a salary schedule that rewards post-graduate educational credentials is not an efficient use of scarce resources.

A final pattern to notice is that the percentage of the variation in student achievement explained by model 4b (20.6 percent)—a random effects model that includes predictors for teacher experience and educational credentials—is much lower than the percentage of variation explained by model 2c (35.1 percent), a model that includes fixed effects for teachers. The explanation is that, while educational credentials and years of teaching experience do indeed predict teacher quality, they explain only a modest part of the variation in teachers' effectiveness in enhancing student achievement.

VI. Lessons

We conclude by describing four lessons about the consequences of initiatives by the Bogotá government to greatly expand access to elementary school education. First, it is likely that expanding access led to a decline in the average quality of education. One mechanism through which this took place is that class sizes in public school classrooms increased. Another is that private schools hired teachers with relatively low levels of experience and credentials (and thus effectiveness). The likely decline in average quality does not mean that expanding access was a mistake—only that, given severe resource constraints, it was not accomplished without cost.

The second lesson is that the government's two-pronged strategy for expanding access seems quite efficient, given the assumption that it was

unable politically to change the structure of public school teachers' compensation schedule or prevent its extension to private schools. Public schools, with very experienced and highly educated teachers, were told to serve more students. They responded by having the most skilled teachers teach the largest classes. It was probably less expensive to subsidize private schools that served low-income students than it was to educate the children in public schools. While schools in the two sectors produced achievement with quite different input mixes, on average, they were equally effective in producing student achievement.

The third lesson concerns the teacher salary schedule. Our evidence indicates that teachers with education beyond a 4-year college degree were not more effective in producing student achievement than teachers with a 4-year degree were. Thus, unless it could be demonstrated that the most highly educated teachers made contributions greater than other teachers in respects not measured by test scores, challenging the structure of the salary scale may be a way to improve efficiency.

Finally, the importance of peer group composition in influencing student achievement creates a difficult public policy problem. Peer group effects create strong incentives for parents with resources to try to place their children in schools with advantaged peers and for schools interested in attracting students from such families to specialize in serving such families. The resulting sorting by SES widens inequality in access to good schooling. Evidence from New Zealand and Chile demonstrates that such sorting by SES does take place in large-scale voucher plans in which the value of the voucher is the same for all students.⁶³

One way to reduce the likelihood of sorting by SES is to use differentially valued vouchers that create incentives for schools to attract students whom they would otherwise spurn.⁶⁴ This idea has a long history.⁶⁵ To date, there has been no significant trial of a school choice system with differentially valued vouchers.

The value of vouchers is only one of many critical details in the design of a voucher system. Others concern whether schools can charge families more than the value of a child's voucher, the quality of information available on school performance, the process through which spaces are allocated in schools for which demand exceeds supply, rules about dismissing difficult students, and the availability of free transportation. By influencing the options available to low-income families and the attractiveness of participation to

⁶³ On New Zealand, see Fiske and Ladd, *When Schools Compete*; on Chile, see Chang-Tai Hsieh and Miguel Urquiola, "When Schools Compete, How Do They Compete?"

⁶⁴ Caroline M. Hoxby, "Ideal Vouchers," unpublished manuscript (Department of Economics, Harvard University, Cambridge, MA, 2001); Dennis Epple and Richard Romano, "Education Vouchers and Cream-Skimming," NBER Working Paper no. 9354 (National Bureau of Economic Research, Cambridge, MA, 2002).

⁶⁵ See the discussion in Murnane, "The Role of Markets in American K-12."

potential suppliers of school services, these details are likely to influence how an educational voucher system would influence the quality of education provided to children from different backgrounds.

Perhaps the most important lesson from this article for the debate over school vouchers is that the concept of unregulated markets is of little value in predicting outcomes. The reason is that, in the language of economics, peer group effects are externalities, and unregulated, competitive markets do not produce an efficient allocation of resources when externalities are present. Any school choice system will be a system with regulations. It is important to pay attention to how, in the face of peer group effects, parents choosing schools and entrepreneurs with different motivations for starting schools respond to the detailed regulations that define a choice system. These responses will determine how the system influences the quality of education provided to children from different backgrounds.

Analytic Appendix

Random Effects Models

In education, students are typically grouped into larger units, such as classrooms and schools. Given their shared experiences, residuals associated with students in the same classroom and/or school are unlikely to be independent of one another, as is required by ordinary least squares regression analysis. Consequently, if this clustering of students within higher-level units is not taken into account, the standard errors associated with the regression parameters will be underestimated.

Multilevel models are a class of regression models that account for the clustering of subjects within larger groups—and hence provide appropriate standard errors. In educational research, one popular approach to multilevel modeling has been introduced by Raudenbush and Bryk (R&B).⁶⁶ An example of the R&B multilevel specification that accounts for the clustering of children within classrooms specifies outcome Y as a function of generic predictor X at both the within-classroom and between-classroom levels, in the following way: Level 1 model (within classroom):

$$Y_{ij} = \beta_{0j} + \beta_{1j}X_{ij} + \varepsilon_{ij}. \quad (\text{A1})$$

Level 2 model (between classroom):

$$\beta_{0j} = \gamma_{00} + u_{0j}$$

$$\beta_{1j} = \gamma_{10} + u_{1j},$$

where i denotes student, j denotes classroom, ε_{ij} is a level 1 residual, and u_{0j} and u_{1j} are level 2 residuals. Statistically, the ε_{ij} , u_{0j} , and u_{1j} residuals are referred to as “random effects.”

Notice that the R&B specification in equation (A1) permits the level 1 re-

⁶⁶ Raudenbush and Bryk, *Hierarchical Linear Models*.

TABLE A1
DESCRIPTION OF OUTCOME, PREDICTOR, AND CONTROL VARIABLES

Variable	Definition
Outcome variable: MSCORE	Student mathematics achievement score on a standardized test administered at the end of fifth-grade (possible score range = 0 to 492 points)
Student-level control predictors:	
FORM	Dichotomous variable indicating test form (0 = form B, 1 = form A)
PRETEST	Student achievement on a mathematics and language test administered at the end of fourth grade (possible score range = 0 to 306 points)
AGE	Student age at the end of the fifth-grade school year (years)
FEMALE	Dichotomous variable indicating student's gender (0 = male, 1 = female)
MOTHED	Ordinal variable indicating mother's educational level (0 = no education, 1 = incomplete primary, 2 = complete primary, 3 = incomplete secondary, 4 = complete secondary and beyond)
MHH	Dichotomous variable indicating whether the student's mother is the head of the household (0 = no, 1 = yes)
DAYABS	Number of days that the student was absent from school in the last 2 weeks
HWHELP	System of dummy variables that indicate the frequency with which the student receives help with homework at home (1 = never, 2 = sometimes, 3 = always). IN our analyses, HWHELP = 3 (always receives help) acts as the reference category
BOOKS	PCA-based composite measure constructed from multiple indicators of the availability of books, storybooks, and encyclopedias in a student's home. Standardized to a mean of zero and a standard deviation of unity, with positive scores indicating better availability
Classroom peer group composition and class size:	
CLMOTHED	"Leave-out" classroom average level of students' mother's education
CLMEAN	"Leave-out" classroom mean score on the pretest
CLASSIZE	Continuous variable indicating number of students enrolled in the classroom
Teacher characteristics:	
EDLEVEL	Ordinal variable indicating the teacher's maximum educational level (either obtained or in process of obtaining). In our analyses, we converted this variable into a system of dichotomous (dummy) predictors, EDLEVEL1 through EDLEVEL5 (1 = pedagogical high school, 2 = educational technology, 3 = licensure in education, 4 = postgraduate courses including master's degree or specialization in education, 5 = other professional qualification). EDLEVEL3 (licensure) was omitted from our analyses as a reference category
TEAEXP	Continuous variable indicating the total number of years that the teacher has been in the teaching profession
SSCALE	Ordinal variable indicating the teacher's step on the salary scale (1–14)
School characteristics:	
PRIVATE	Dichotomous variable indicating whether the school is public or private (1 = private, 0 = public)

gression parameters β_{0j} and β_{1j} (which represent the within-classroom relationship between Y and X) to vary randomly across classrooms at level 2. The model, therefore, is often referred to as a "random intercepts, random slopes" model. In this model the u_{0j} are the "random effects" of the level 1 intercept, and the u_{1j} are the random effects of the level 1 slope, respectively. Consequently, when there are no predictors in the model at level 2, $\text{var}(u_{0j})$ is the variation in classroom-level in-

TABLE A2
ESTIMATED REGRESSION COEFFICIENTS ASSOCIATED WITH CONTROL PREDICTORS, BY MODEL

	Model 1	Model 2a	Model 2b	Model 2c	Model 2a (r.e.)	Model 2b (r.e.)	Model 2c (r.e.)	
Constant					296.027***	295.945***	296.091***	
FORM	−6.28***	−6.328***	−6.335***	−6.334***	−6.440***	−6.403***	−6.457***	
PRETEST	.308***	.302***	.302***	.303***	.329***	.314***	.310***	
PRETEST SQ	.001***	.001***	.001***	.001***	.001***	.001***	.001***	
FEMALE	−8.17***	−8.160***	−8.145***	−8.140***	−7.358***	−8.125***	−8.160***	
AGE	−2.73***	−2.579***	−2.649***	−2.593***	−2.160**	−2.311**	−2.214**	
AGE SQ	−.84*	−.906**	−.885**	−.897**	−1.003**	−.840*	−.869*	
DAYABS	−1.85***	−1.807***	−1.762***	−1.805***	−1.939***	−1.888***	−1.832***	
MHH	4.63***	4.385***	4.395***	4.399***	4.521***	4.730***	4.702***	
BOOKS	2.39***	2.453***	2.465***	2.446***	2.693***	2.363***	2.404***	
HWHELP1	7.73**	7.792**	7.672**	7.745**	7.650**	7.695**	7.626**	
HWHELP2	7.60***	7.592***	7.528***	7.526***	7.374***	7.503***	7.463***	
MOTHEd	.512	.579	.936	.796	1.868**	.991	.974	
	Model 3a*	Model 3a	Model 3b	Model 3c	Model 4a	Model 4b	Model 4c	Model 4d
Constant	292.310***	295.540***	298.140***	300.424***	292.056***	277.394***	295.328***	293.681***
FORM	−6.782***	−6.447***	−6.401***	−6.359***	−6.353***	−6.351***	−6.389***	−6.392***
PRETEST		.329***	.318***	.313***	.311***	.309***	.313***	.312***
PRETEST SQ		.001***	.001***	.001***	.001***	.001***	.001***	.001***
FEMALE		−7.307***	−8.094***	−7.665***	−7.809***	−7.882***	−8.512***	−8.524***
AGE		−2.134**	−2.343***	−2.517***	−2.570***	−2.561***	−2.417***	−2.400***
AGE SQ		−1.005**	−.873**	−.922**	−.898**	−.883**	−.827*	−.826*
DAYABS		−1.938***	−1.896***	−1.866***	−1.860***	−1.865***	−1.894***	−1.899***
MHH		4.517***	4.660***	4.413***	4.469***	4.426***	4.801***	4.824***
BOOKS		2.683***	2.456***	2.460***	2.442***	2.405***	2.388***	2.381***
HWHELP1		7.603**	7.854**	7.697**	7.707**	7.662**	8.044**	8.024**
HWHELP2		7.348***	7.577***	7.531***	7.568***	7.544***	7.712***	7.695***
MOTHEd		1.850*	1.244 ⁺	1.044	.998	.899	1.055	1.042

⁺ $p < .10$.

* $p < .05$.

** $p < .01$.

*** $p < .001$.

TABLE A3
TAXONOMY OF CLASSROOM RANDOM EFFECTS REGRESSION MODELS ESTIMATING THE EFFECTS OF
SCHOOL SECTOR ON STUDENTS' FIFTH-GRADE MATHEMATICS ACHIEVEMENT SCORE

	3a* (No Student-Level Controls)	3a	3b (Coefficient on CLMOTHEd Con- strained to Its Value in Model 2b)	3c (Coefficients on Class Size and CLMOTHEd Con- strained to Their Values in Model 2c)
Class-level predictors:				
CLMOTHEd			(12.199)	(7.932)
CLASSIZE				(23.919)
School sector (PRIVATE = 1):				
PRIVATE	8.239*	.754	-3.789 ⁺	-14.502***
Goodness of fit:				
R^2 within	.009	.200	.201	.201
R^2 between	.050	.650	.681	.294
R^2 overall	.025	.290	.316	.195

NOTE.— n students = 3,095, n classrooms = 97, n teachers = 77. Regression coefficients for control variables can be found in app. table A2.

⁺ $p < .10$.

* $p < .05$.

*** $p < .001$.

tercepts across classrooms, and $\text{var}(u_{1j})$ is the variation in the classroom-level slopes across classrooms ($\text{var}(\varepsilon_{ij})$ is the “within-classroom” residual variation in Y). Models like equation (A1) can be estimated using the HLM software developed and marketed by R&B.

However, the R&B “level 1/level 2” specification is not the only way to specify a multilevel model, nor the only way to deal with the clustering of individuals within larger units in social data. Notice that we can collapse the R&B level 1/level 2 specification into a single composite regression model by substituting for parameters β_{0j} and β_{1j} from R&B’s level 2 model into their level 1 model, and reorganizing terms, as follows:

$$Y_{ij} = \gamma_{00} + \gamma_{10}X_{ij} + (u_{0j} + u_{1j}X_{ij} + \varepsilon_{ij}). \quad (\text{A2})$$

We recognize equation (A2) as a regression model whose structural part contains the usual parameters—an intercept γ_{00} and a slope γ_{10} —and whose stochastic (composite residual) part has a complex form that is heteroskedastic (it depends on predictor X) and autocorrelated across children within classrooms (as a result of the shared dependence on the u ’s whose (random) values are common across children within classroom). Such models can be fitted by any computer software that provides generalized least squares regression analysis. And, if such software is employed, it generally provides estimates that are identical, or very close, to the estimates provided by the HLM software.

We emphasize that adopting the standard R&B level 1/level 2 specification leads to a particular hypothesized form for the composite residual error covariance structure. However, these specifications for the structure of the stochastic part of the R&B model may not be met by all data sets. For instance, depending on the problem being investigated, the error structure hypothesized by the standard R&B model may be too simple—in which case, more complex error covariance structures must be hypothesized. Many software programs—such as SAS PROC MIXED—

provide for such complexity. Alternatively, the error covariance structure implicit in the standard R&B model may be more complex than is really needed—that is, over-specified. In either case, sensible testing—with standard tools—can provide insight into the appropriate functional form of the stochastic part of the composite multilevel model.

One common simplification of the R&B model is to eliminate the heteroskedasticity implicit in the composite residual by the dependence on predictor X , as follows:

$$Y_{ij} = \gamma_{00} + \gamma_{10}X_{ij} + (u_{0j} + \varepsilon_{ij}). \quad (\text{A3})$$

This corresponds to the following R&B level 1/level 2 specification: level 1 model (within classroom):

$$Y_{ij} = \beta_{0j} + \beta_{1j}X_{ij} + \varepsilon_{ij}, \quad (\text{A4})$$

and level 2 model (between classrooms):

$$\begin{aligned} \beta_{0j} &= \gamma_{00} + u_{0j} \\ \beta_{1j} &= \gamma_{10}. \end{aligned}$$

Notice that, in equations (A3) and (A4), the formerly hypothesized random variation of slopes across classrooms has been set to zero. Formally, the model in equations (A3) and (A4) is referred to as a “random intercepts” model. In our work, preliminary tests revealed that a “random intercepts” model was adequate and preferred to a “random intercepts, random slopes” model. Hence, it is the “random intercepts” model that we fit throughout our analyses and, consistent with the usual practices of econometrics, we refer to this “random intercepts” model as a “random effects” model. We fit it to our data using the “xtreg” procedure in STATA. Providing that its assumptions are met (and, in our work, tests indicate that they are), the “random effects” specification represents our data-structure more parsimoniously. It therefore properly—and more powerfully—accounts for the clustering of children within classrooms.

Fixed Effects Models

As explained above, the standard “random intercepts” model can parsimoniously and powerfully account for the clustering of children within classrooms by using a random effects specification for the within-classroom intercepts. However, a simple modification of the random effects specification provides an even more robust representation of the clustering of children within classrooms.

We achieve this modification by treating the same u_{0j} s that account for the clustering of children within classrooms in equation (A4) not as “random effects” but as “fixed effects.” This latter approach to multilevel modeling has traditionally

been preferred by economists because of its advantageous properties.⁶⁷ Specifically, the u_{0j} s that represent unobserved between-classroom differences in the outcome can be moved from the residual term in the composite model and treated as part of the structural part of the regression model, as follows:

$$Y_{ij} = (\gamma_{00} + u_{0j}) + \gamma_{10}X_{ij} + \varepsilon_{ij}. \quad (\text{A5})$$

Notice that the unobserved effects of classrooms are now grouped with the overall intercept in equation (A5), essentially requiring that each classroom has its own intercept. In practice, the analysis can be conducted by introducing a set of dummy predictors into the model, one per classroom, and the coefficients associated with these dummies are the required fixed effects. Furthermore, the model can now be fitted by OLS methods as the remaining residual term—the ε_{ij} satisfies the usual OLS assumptions. It is this latter specification that we refer to as the “fixed effects” model in our analysis. And, again, whether the effects of classrooms are treated as “random” or “fixed” effects, the model accounts appropriately for the clustering of children within classrooms.

If both the fixed effects and random effects specifications for classroom effects reasonably account for the clustering of children within classrooms, then how do we choose one approach over the other? Clearly, each approach must have its own advantages and disadvantages. One way in which the fixed effects approach is disadvantaged, for instance, is that many additional dummy predictors have been introduced into the structural part of the regression model to represent classrooms, and so there are many additional associated regression parameters to be estimated, with an appropriate sacrifice of degrees of freedom and statistical power.

The fixed effects approach is clearly superior to the random effects approach in one important way. By using the dummy predictor specification for classrooms, the fixed effects approach accounts for all possible observed and unobserved differences among classrooms, and it does not matter if these differences are correlated with predictors at other levels, such as X , that may have been included in the model (because predictors in a regression model may naturally be correlated). However, if the u_{0j} are treated as random effects but are correlated with other predictors in the model, then estimates of the effect of predictor X on Y will be biased because a correlation between predictors and residuals is not permitted in any kind of regression model.

Thus, in preferring a fixed effects approach to a random effects approach, we can sacrifice a few degrees of freedom for a potential reduction in bias. This is an important and intelligent trade-off that can be advantageous in our data set. In fitting teacher fixed effects models, for instance, we can account for all possible observed and unobserved differences in the outcome among teachers but still preserve some variation in the outcome due to classrooms (because there are

⁶⁷ Fixed effects models are often used with panel data. However, they are also commonly used with cross-sectional data. Julian R. Betts (“Does School Quality Matter? Evidence from the National Longitudinal Survey of Youth,” *Review of Economics and Statistics* 77 [May 1995]: 231–50) incorporates fixed effects for states within the United States. Patrick J. McEwan (“Peer Effects on Student Achievement”) uses school, twin, and sibling fixed effects. Jere R. Behrman and Barbara L. Wolfe (“Does More Schooling Make Women Better Nourished and Healthier? Adult Sibling Random and Fixed Effects Estimates for Nicaragua,” *Journal of Human Resources* 24 [Autumn 1989]: 644–63) use both sibling fixed effects and sibling random effects models.

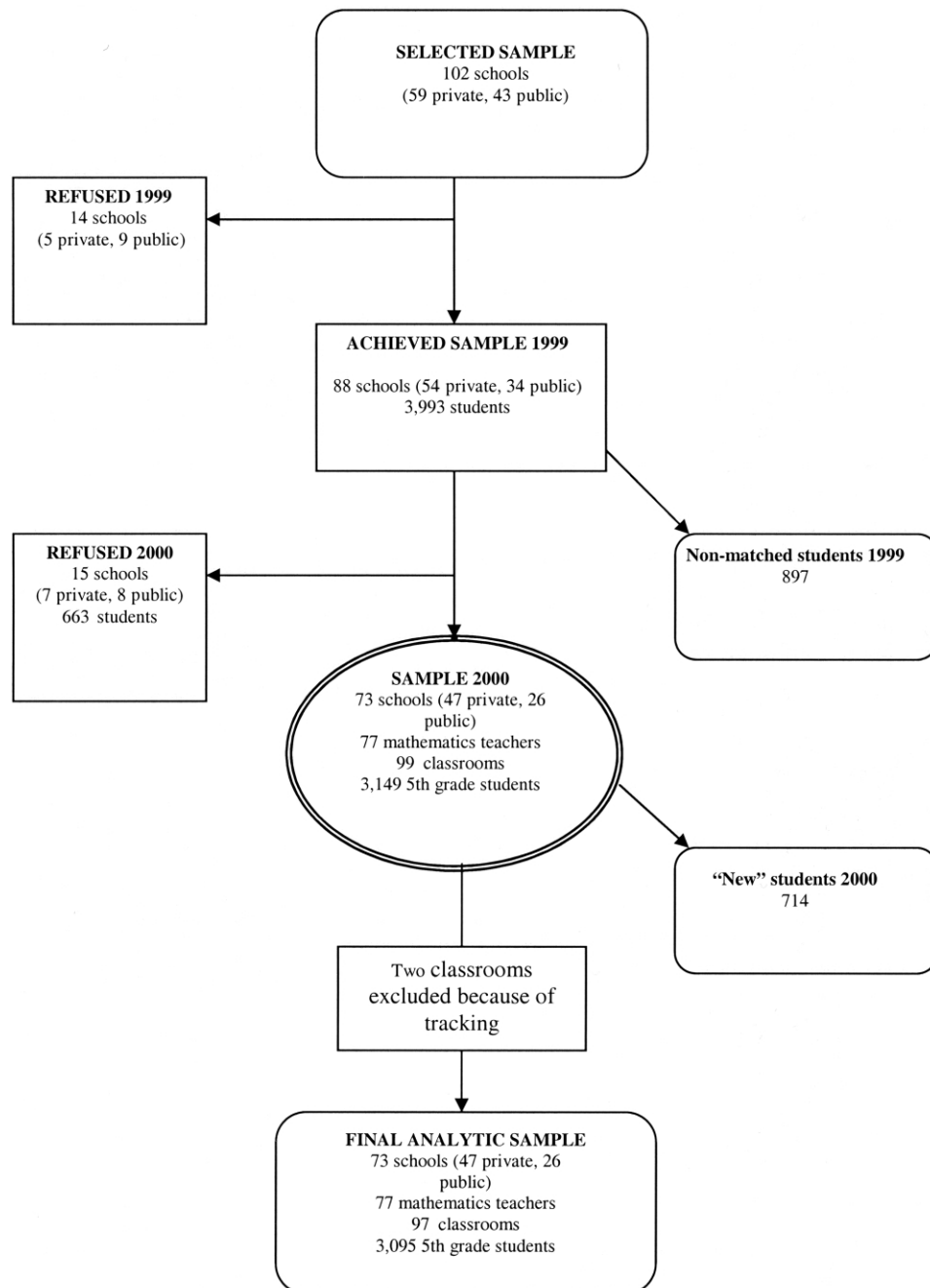


FIG. A1.—Sampling process for the two-wave data collection

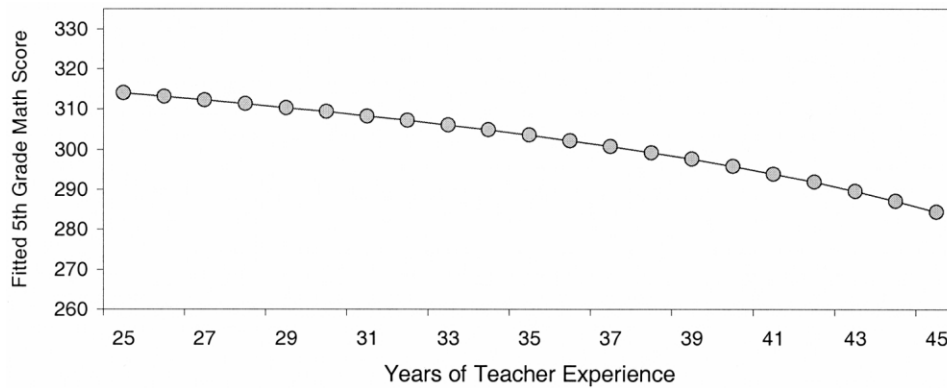


FIG. A2.—Plot displaying the fitted relationship between students' fifth-grade mathematics achievement score and class size, for a prototypical student in a classroom of average peer group SES (based on model 2c; n students = 3,095, n classrooms = 97, n teachers = 77).

teachers present in our sample who teach more than one classroom). The remaining classroom-level outcome variation can then be predicted by important classroom level features, such as the class size and peer group composition, without worrying about the potentially biasing impact of any unobserved teacher effects.

Unfortunately, also in our data set, once we have controlled for the teacher fixed effects, there is relatively little variation in the outcome remaining to be predicted by the class size and peer group composition of the classrooms taught by the same teacher. Therefore, although estimates of the associated regression parameters associated with these predictors are less biased, their standard errors are larger and the corresponding null hypotheses less easy to reject. In our analyses, we resolve the pros and cons of the choice between the fixed effects and the random effects approach by implementing both types of strategy (again, using the STATA “xtreg”) and by using the one to inform the other. For instance, in fitting model 2, we estimate a teacher fixed effects model, in order to obtain an unbiased estimate of the effects of class size and peer group composition. In some cases, these estimates are large but not statistically significant, due to a lack of power. So, we also estimate a classroom random effects model and compare the estimates from this model to the ones obtained from the fixed effects model. If the estimates from the former model are similar to those from the latter model, then we can conclude that the random effects model provides unbiased estimates after all. If such is the case, then we can rely on the (more precise) standard errors from the random effects model to make statistical inferences. For example, the peer group effects estimate in model 2b is statistically different from zero in the random effects model, but not in the fixed effects model. However, because the estimates in the fixed effects and random effects models are almost identical in size, we have reason to believe that the estimate from either model is reasonable and unbiased. In contrast, the estimated effect of class size from the fixed effects model is very different than

the one from the random effects model. In this instance, we conclude that the random effects estimate is indeed biased.⁶⁸

Next, we fit models 3 and 4 to investigate private school effectiveness, as well as the contribution of teacher characteristics to student achievement. Because there is no variation in school sector and teacher characteristics across classrooms taught by the same teacher, we can no longer use a teacher fixed effects model, and instead use a random effects model. Unfortunately, the estimates from random effects models are likely to be biased. We therefore again adopt a “mixed” approach. As described above, the class size and peer effects estimates from the fixed effects regressions (model 2) are the best ones available (i.e., the least biased). In our random effects models, we therefore constrain the effects of class size and peer group composition to take on their values from the fixed effects models. This approach reduces potential bias in the estimates of the other regressors in models 3 and 4.

⁶⁸ We used a Hausman specification test to evaluate whether estimates from the fixed effects and random effects models differed based on model 2c and find that they do at all standard levels of significance. This indicates that estimates from the random effects model are indeed biased.